Exact values

1 Consider the equation $\sin\theta = -\frac{1}{2}$. State the number of solutions for θ, in the domain $0 < \theta < \frac{\pi}{2}$.

2 For each of the following equations, find the exact value of acute angle x, in the domain $0 \leq x \leq \frac{\pi}{2}$:

a $\tan x = 1$ **b** $\sin x = \frac{\sqrt{3}}{2}$ **c** $\cos x = \frac{1}{\sqrt{2}}$ **d** $2\tan x = \frac{2\sqrt{3}}{3}$

3 For each of the following, find the exact values of x, where $-\pi < x < \pi$:

a $\tan x = \sqrt{3}$

b $\tan x = \frac{1}{\sqrt{3}}$

c $\sin 2x = \frac{1}{2}$

d $\tan\left(x - \frac{2\pi}{3}\right) = 0$

4 For each of the following, find the exact values of θ, in the given domain:

a $\tan\theta = 0$, where $-\pi \leq \theta \leq \pi$

b $\sin\theta = 0$, where $-4\pi \leq \theta \leq 4\pi$

c $12\cos\theta - 6\sqrt{3} = 0$, where $-\pi < \theta < \pi$

d $4\sin\theta - 3 = -5$, where $-2\pi \leq \theta \leq 2\pi$

e $\cos^2\left(\frac{\theta}{2}\right) - 1 = 0$, where $-2\pi \leq \theta \leq 2\pi$

f $4\sin^2(\theta) = 1$, where $-4\pi < \theta < 4\pi$

5 For each of the following, find the exact values of x, over the domain $[0, 2\pi)$:

a $\sin x = \frac{1}{2}$

b $\cos x = -\frac{\sqrt{3}}{2}$

c $6\cos x + 2 = -1$

d $6\cos x - 3\sqrt{2} = 0$

e $2\tan x + 3 = 5$

f $2\sin x + 2 = 1$

g $\cos x \tan x = \cos x$

h $\tan^2(x) + 2\tan x + 1 = 0$

i $2\sin 3x - \sqrt{2} = 0$

j $\sin\left(\frac{x}{2}\right) = \frac{\sqrt{3}}{2}$

k $2\sin\left(\frac{x}{2}\right) = \sqrt{3}$

l $\sin\left(\frac{x}{2}\right) = -\cos\left(\frac{x}{2}\right)$

m $\cos\left(\frac{x}{2}\right) = 1 - \cos\left(\frac{x}{2}\right)$

n $\sin\left(\frac{x}{2}\right) = 1 - \sin\left(\frac{x}{2}\right)$

o $\cos^2\left(\frac{x}{2}\right) - 1 = 0$

p $\sin^2\left(\frac{x}{2}\right) - 1 = 0$

q $36(1 - \cos x)(1 + \cos x) = 27$

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6 For each of the following, find the exact values of x, over the domain $0 \leq x \leq 2\pi$:

a $\sin x = 1$

b $\cos x = -1$

c $\sin x = \frac{1}{2}$

d $\sin x = -\frac{1}{\sqrt{2}}$

e $\cos x = -\frac{1}{2}$

f $\tan x = -\frac{1}{\sqrt{3}}$

g $10 \sin x - 5\sqrt{3} = 0$

h $4 \cos x - 2 = 0$

i $2\sqrt{3} \tan x - 2 = 0$

j $\sin 2x = \frac{1}{\sqrt{2}}$

k $\tan 3x = -\frac{1}{\sqrt{3}}$

l $\sin\left(x + \frac{\pi}{6}\right) = \frac{1}{\sqrt{2}}$

m $\cos\left(x + \frac{\pi}{6}\right) = -\frac{1}{2}$

n $\sin x = -\cos x$

o $\sin\left(x + \frac{\pi}{3}\right) = \frac{1}{2}$

p $\sqrt{3} \tan\left(\frac{x}{2}\right) = -3$

q $\cos\left(x + \frac{\pi}{4}\right) = -\frac{1}{\sqrt{2}}$

r $\sin\left(x - \frac{4\pi}{3}\right) = \frac{1}{2}$

7 Find the exact solutions of the following equations where $0 \leq \theta \leq 2\pi$:

a $\cos^2 \theta - 5 \cos \theta + 4 = 0$

b $2 \sin^2 \theta + \sin \theta - 1 = 0$

c $\tan^2 \theta + \sqrt{3} \tan \theta = 0$

d $\sin^2 \theta - \cos^2 \theta = 0$

Other radian values

8 Consider the equation $\cos \theta = 0.42$. State the number of solutions for θ over the domain $0 < \theta < 2\pi$.

9 For each of the following equations:

i Find the acute angle θ in radians that solves the equation, correct to two decimal places.

ii Hence find all solutions to θ over the domain $[0, 2\pi]$, correct to two decimal places.

a $\sin \theta = 0.3420$

b $\cos \theta = 0.9063$

c $\tan \theta = 0.8340$

d $\cos \theta = 0.2345$

e $\sin \theta = 0.9921$

f $\tan \theta = 0.7743$

g $\cos \theta = 0.67$

h $\sin \theta = 0.091$

i $\cos 2\theta = 0.9$

10 Jessica finds the acute angle solution to the equation $\sin x = k$, is $x = p$.
Find another solution for the equation over the domain $0 < x < 2\pi$, in terms of p.

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11 Toby finds the acute angle solution to the equation $\cos x = m$, is $x = n$.
Find another solution for the equation over the domain $0 < x < 2\pi$, in terms of n.

12 Sarah finds the acute angle solution to the equation $\tan x = h$, is $x = t$.
Find another solution for the equation over the domain $0 < x < 2\pi$, in terms of t.

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